

NCERT Exemplar Solutions For Class 12

Physics

Chapter 11 - Dual Nature of Radiation and Matter

Multiple Choice Questions

1. A particle is dropped from a height H . The de Broglie wavelength of the particle as a function of height is proportional to

(a) H

(b) $H^{\frac{1}{2}}$

(c) H^0

(d) $H^{-\frac{1}{2}}$

Ans: When the body is falling freely from a height, the velocity is

$$H = v$$

$$= \sqrt{2gH}$$

according to de-Broglie wavelength

$$\lambda = \frac{h}{p}$$

$$\lambda = \frac{h}{mv}$$

$$= \frac{h}{m\sqrt{2gH}}$$

Where,

h , m and g are constant.

$$\therefore \frac{h}{m\sqrt{2g}} \text{ is constant}$$

$$\Rightarrow \lambda \propto \frac{1}{\sqrt{H}}$$

$$\Rightarrow \lambda \propto H^{-\frac{1}{2}}$$

Thus, option (d) is correct.

2. The wavelength of a photon needed to remove a proton from a nucleus which is bound to the nucleus with 1 MeV energy is nearly

(a) 1.2 nm

(b) 1.2×10^{-3} nm

(c) 1.2×10^{-6} nm

(d) 1.2×10^1 nm

Ans: we know that the energy of the photon should be equal to the binding energy of proton

Thus, energy of photon = 1 MeV

$$= 10^6 \times 1.6 \times 10^{-19} \text{ J}$$

$$\lambda = \frac{hc}{E}$$

$$\Rightarrow \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{1.6 \times 10^{-13}}$$

$$\Rightarrow \frac{6.63 \times 3}{1.60} \times 10^{-26+13}$$

$$\Rightarrow \frac{19.89}{1.60} \times 10^{-13}$$

$$= 12.4 \times 10^1 \times 10^{-13}$$

$$\Rightarrow 1.24 \times 10^{-9} \times 10^{-3}$$

$$= 1.24 \times 10^{-3} \text{ nm}$$

So, option (b) is correct.

3. Consider a beam of electrons (each electron with energy E_0) incident on a metal surface kept in an evacuated chamber. Then

- (a) no electrons will be emitted as only photons can emit electrons.**
- (b) electrons can be emitted but all with an energy, E_0 .**
- (c) electrons can be emitted with any energy, with a maximum of $E_0 - \phi$ (ϕ is the work function).**
- (d) electrons can be emitted with any energy, with a maximum of E_0 .**

Ans: When a beam of electrons in which each electron of energy E_0 incident on the metal surface which is kept in vacuum then because of elastic collisions with electrons on the surface, the energy of incident electrons will be transferred to the emitted electrons. For the emission of electrons below the surface the part of E_0 of incident electrons is consumed against work function so the energy of emitted electrons becomes less than E_0 . Thus, the maximum energy of emitted electrons can be E_0 . So, the option (d) is correct.

4. Consider Fig. 11.7 in the NCERT text book of physics for Class XII. Suppose the voltage applied to A is increased. The diffracted beam will have the maximum at a value of θ that

(a) will be larger than the earlier value.

(b) will be the same as the earlier value.

(c) will be less than the earlier value.

(d) will depend on the target

Ans:

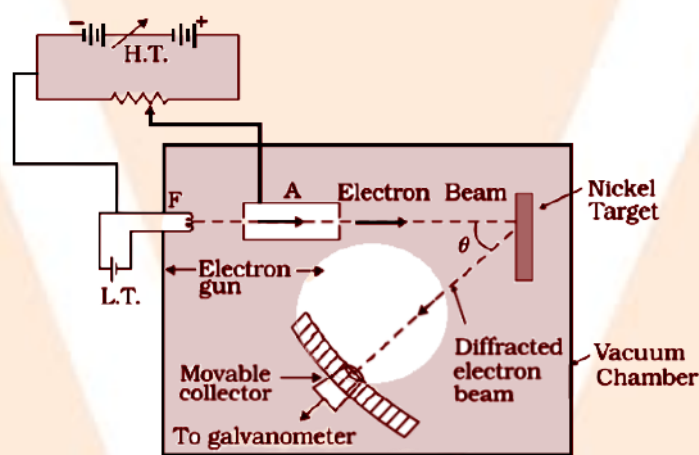


FIGURE 11.7 Davisson-Germer electron diffraction arrangement.

Through Davisson-Germer experiment, we know that diffracted beam of electrons

has the de-Broglie wavelength

$$\lambda_d = \frac{12.27}{\sqrt{V}} \text{ \AA} \quad \dots(i)$$

Here, V is the voltage applied. When there is a maximum of the diffracted electrons at an angle θ , then

$$2d \sin \theta = \lambda$$

From equation (i), as there are increases in the applied voltage in this experiment, the wavelength λ_d decreases in turn $\sin \theta$ or θ decreases by relation (ii). Thus, this verifies the option (c).

5. A proton, a neutron, an electron and an α -particle have the same energy. Then their de Broglie wavelengths compare as

(a) $\lambda_p = \lambda_n > \lambda_e > \lambda_\alpha$

(b) $\lambda_\alpha < \lambda_p = \lambda_n > \lambda_e$

(c) $\lambda_e < \lambda_p = \lambda_n > \lambda_\alpha$

(d) $\lambda_e = \lambda_p = \lambda_n = \lambda_\alpha$

Ans: According to de-Broglie wavelength

$$\lambda_d = \frac{h}{p}$$

$$E_p = E_n = E_e = E_\alpha$$

$$K.E. = K = \frac{1}{2}mv^2$$

$$\Rightarrow 2K = mv^2$$

$$\Rightarrow 2Km = m^2v^2$$

$$\Rightarrow 2mK = p^2$$

$$\Rightarrow \sqrt{2mK} = p$$

$$\therefore \lambda_d = \frac{h}{p}$$

$$\Rightarrow \lambda_d = \frac{h}{\sqrt{2mK}}$$

$$\text{Or } \lambda_d = \frac{h}{\sqrt{2mK}}$$

Where, h and E (K.E.) is constant.

$$\therefore \lambda \propto \frac{1}{\sqrt{m}}$$

Thus,

$$m_{\alpha} > m_p = m_n > m_e$$

So, $\lambda_{\alpha} < \lambda_p = \lambda_n > \lambda_e$, option (b) is correct.

6. An electron is moving with an initial velocity $v = v_0 \hat{i}$ and is in a magnetic field $B = B_0 \hat{j}$. Then it's de Broglie wavelength

- (a) remains constant.**
- (b) increases with time.**
- (c) decreases with time.**
- (d) increases and decreases periodically.**

Ans: In the question, it is given that

$$v = v_0 \hat{i}$$

$$\text{And, } B = B_0 \hat{j}$$

The force on moving electron is perpendicular magnetic field B is,

$$F = -e(\vec{v} \times \vec{B})$$

$$= -e[v_0 \hat{i} \times B_0 \hat{j}]$$

$$= -ev_0 B_0 \hat{i} \times \hat{j}$$

$$F = -ev_0 B_0 \hat{k}$$

So, here the force is perpendicular to both v and B as it is perpendicular to the velocity. Thus, there will no change in v or mv so the de-Broglie wavelength remains the same.

7. An electron (mass m) with an initial velocity $v = v_0 \hat{i}$ ($v_0 > 0$) is in an electric field $E = -E_0 \hat{i}$ ($E_0 = \text{constant} > 0$). It's de-Broglie wavelength at time t is given by

(a) $\frac{\lambda}{\left[1 + \frac{eE_0 t}{m v_0}\right]}$

(b) $\lambda_0 \left[1 + \frac{eE_0}{m v_0} t\right]$

(c) λ_0

(d) $\lambda_0 t$

Ans: at initial, de-Broglie wavelength

$$\lambda_0 = \frac{h}{m v_0}$$

Force acting on electron

$$F = qE$$

$$F = (-e)(-E_0 \hat{i})$$

$$ma = eE_0 \hat{i}$$

$$a = \frac{eE_0}{m} \hat{i}$$

After time t , velocity of electron is

$$v = v_0 + at$$

$$v = v_0 i + \frac{eE_0}{m} i \cdot t$$

$$v = \left[v_0 + \frac{eE_0 t}{m} \right] \hat{i}$$

$$\therefore \text{New de-Broglie wavelength } \lambda = \frac{h}{mv}$$

$$\lambda = \frac{h}{m \left[v_0 + \frac{eE_0 t}{m} \right] \hat{i}}$$

$$= \frac{h}{mv_0 \left[1 + \frac{eE_0 t}{mv_0} \right]}$$

$$\lambda = \frac{\lambda_0}{\left[1 + \frac{eE_0 t}{mv_0} \right]} \left(\because \frac{h}{mv_0} = \lambda_0 \text{ from eqn.I} \right)$$

8. An electron (mass m) with an initial velocity $v = v_0 \hat{i}$ is in an electric field

$E = E_0 \hat{j}$. If, $\lambda_0 = \frac{h}{mv_0}$, it's de-Broglie wavelength at time t is given by

(a) λ_0

(b) $\lambda_0 \sqrt{1 + \frac{e^2 E_0^2 t^2}{m^2 v_0^2}}$

(c) $\frac{\lambda_0}{\sqrt{1 + \frac{e^2 E_0^2 t^2}{m^2 v_0^2}}}$

(d) $\frac{\lambda_0}{\left[1 + \frac{e^2 E_0^2 t^2}{m^2 v_0^2} \right]}$

Ans: We know that de-Broglie wavelength

$$\lambda_0 = \frac{h}{mv_0}$$

Force acting on the moving electron due to electric field E

$$F = -eE$$

$$\Rightarrow F = -eE_0 \hat{j}$$

Because of electric field, acceleration in electron due to force,

$$ma = -eE_0 \hat{j}$$

$$\Rightarrow a = \frac{-eE_0}{m} \hat{j}$$

And acceleration on electron acts along $-Y$ direction, thus the initial velocity of electron along X - axis

$$v_{x_0} = v_0 \hat{i}$$

Thus, the initial velocity of electron in Y direction is zero

$$\therefore v_{y_0} = 0$$

And the velocity of electron after time t along X - axis

$$v_x = v_0 \hat{i}$$

Thus, velocity of electron after t along Y - axis

$$v = u + at$$

$$v_y = 0 + \left(\frac{-eE_0}{m} \hat{j} \right) t$$

$$v_y = \frac{-eE_0}{m} \hat{j} t$$

After time t , magnitude of velocity of electron is

$$v = \sqrt{v_x^2 + v_y^2}$$

$$v = \sqrt{v_0^2 + \left(\frac{-eE_0}{m} \hat{j}t \right)^2}$$

$$|v| = v_0 \sqrt{1 + e^2 E_0^2 t^2}$$

$$\therefore \lambda' = \frac{h}{mv}$$

$$= \frac{h}{mv_0 \sqrt{1 + \frac{e^2 E_0^2 t^2}{m^2 v_0^2}}}$$

$$v' = \frac{1}{\sqrt{1 + \frac{e^2 E_0^2 t^2}{m^2 v_0^2}} \left(\because \lambda_0 = \frac{h}{mv_0} \right)}$$

Thus, option (c) is verified.

Multiple Choice Questions-II (More than one option)

9. Relativistic corrections become necessary when the expression for the kinetic energy $\frac{1}{2}mv^2$, becomes comparable with mc^2 , where m is the mass of the particle. At what de Broglie wavelength will relativistic corrections become important for an electron?

- (a) $\lambda = 10 \text{ nm}$
- (b) $\lambda = 10^{-1} \text{ nm}$
- (c) $\lambda = 10^{-4} \text{ nm}$
- (d) $\lambda = 10^{-6} \text{ nm}$

Ans: according to de-Broglie wavelength

$$\lambda = \frac{h}{mv}$$

$$\Rightarrow v = \frac{h}{m\lambda}$$

Here, $h = 6.6 \times 10^{-34}$

$$m = 9 \times 10^{-31} \text{ kg}$$

$$= \frac{6.6 \times 10^{-34}}{9 \times 10^{-31} \lambda}$$

$$= \frac{6.6 \times 10^{-34+31}}{9\lambda}$$

$$= \frac{0.73 \times 10^{-3}}{\lambda}$$

For option

$$(a) \lambda = 10 \text{ nm}$$

$$= 10 \times 10^{-9} \text{ m}$$

$$= 10^{-8} \text{ m}$$

$$\therefore v = \frac{7.3 \times 10^{-4}}{10^{-8}}$$

$$= 7.3 \times 10^4$$

$$\approx 10^4 < 3 \times 10^8 \text{ (Speed of light)}$$

$$(b) \lambda = 10^{-1} \text{ nm}$$

$$= 10^{-1} \times 10^{-9} \text{ m}$$

$$= 10^{-10} \text{ m}$$

$$\therefore v = \frac{7.3 \times 10^{-4}}{10^{-10}}$$

$$= 7.3 \times 10^{-4+10}$$

$$= 7.3 \times 10^6$$

$$\approx 10^7 < 10^8 \text{ (Speed of light)}$$

$$(c) \lambda = 10^{-4} \text{ nm}$$

$$= 10^{-4} \times 10^{-9} \text{ m}$$

$$= 10^{-13} \text{ m}$$

$$\therefore v = \frac{7.3 \times 10^{-4}}{10^{-13}}$$

$$= 7.3 \times 10^{-4+13}$$

$$7.3 \times 10^9$$

$$\approx 10^9 > 10^8 \text{ (Speed of light)}$$

$$(d) \lambda = 10^{-6} \text{ nm}$$

$$= 10^{-6} \times 10^{-9} \text{ m}$$

$$= 10^{-15} \text{ m}$$

$$\therefore v = \frac{7.3 \times 10^{-4}}{10^{-15}}$$

$$= 7.3 \times 10^{-4+15}$$

$$\approx 10^{11} > 10^8 \text{ (Speed of light)}$$

Here we can see that the velocity of electron is more for option (c) and (d) where the relativistic correction become necessary although the speed of

electron is $7.3 \times 10^6 \text{ ms}^{-1}$ which is comparable with the speed of light, thus the correct answers of this question is option (c) and (d).

10. Two particles A_1 and A_2 of masses m_1, m_2 ($m_1 > m_2$) have the same de-Broglie wavelength. Then

(a) their momenta are the same.

(b) their energies are the same.

(c) energy of A_1 is less than the energy of A_2 .

(d) energy of A_1 is more than the energy of A_2 .

Ans: We know that de-Broglie wavelength

$$\lambda = \frac{h}{p}$$

$$p = \frac{h}{\lambda}$$

$$\text{or } p \propto \frac{1}{\lambda}$$

$$\frac{p_1}{p_2} = \frac{\lambda_2}{\lambda_1}$$

$$\because \lambda_1 = \lambda_2 = \lambda \text{ (given)}$$

$p_1 = p_2$ this verifies option (a)

$$E_a = \frac{1}{2}mv^2$$

$$= \frac{1}{2} \frac{m^2 v^2}{m}$$

$$= \frac{p^2}{2m}$$

$$E \propto \frac{1}{m} \left[\text{as } p_1 = p_2 = \text{constant (as proved above)} \right]$$

$$\frac{E_1}{E_2} = \frac{m_2}{m_1}$$

$$\frac{E_1}{E_2} < 1 \left[\because m_1 > m_2 \text{ (given)} \right]$$

$\therefore E_2 > E_1$ this verifies answer (c).

11. The de Broglie wavelength of a photon is twice the de-Broglie wavelength of an electron. The speed of the electron is $v_e = \frac{c}{100}$ Then

(a) $\frac{E_e}{E_p} = 10^{-4}$

(b) $\frac{E_e}{E_p} = 10^{-2}$

(c) $\frac{p_e}{m_e c} = 10^{-2}$

(d) $\frac{p_e}{m_e c} = 10^{-4}$

Ans: de-Broglie wavelength

$$\lambda = \frac{h}{p}$$

$$\therefore \lambda_e = \frac{h}{m_e v_e}$$

$$= \frac{h}{m_e \frac{c}{100}}$$

$$= \frac{100h}{m_e c} \dots(i)$$

Now,

$$K.E. = \frac{1}{2}mv^2$$

$$\Rightarrow E = \frac{m^2 v^2}{2m} \dots(ii)$$

$$\Rightarrow m_e v_e = \sqrt{2m_e E_e}$$

Now,

$$\lambda_e = \frac{h}{m_e v_e}$$

$$= \frac{h}{\sqrt{2m_e E_e}} \dots[\text{from (ii)}]$$

$$\Rightarrow E_e = \frac{h^2}{2m_e \lambda_e^2} \dots(iii)$$

For proton

$$\lambda_p = 2\lambda_e \text{ [given]}$$

$$\therefore E_p = \frac{hc}{\lambda_p}$$

$$= \frac{hc}{2\lambda_e}$$

$$\text{Now, } \frac{E_p}{E_e} = \frac{hc}{2\lambda_e} \times \frac{2m_e \lambda_e^2}{h^2}$$

$$= \frac{\lambda_e m_e c}{h} [\text{from (iii)}]$$

$$= \frac{100h}{m_e c} \times \frac{m_e c}{h}$$

$$= 100$$

$$\Rightarrow \frac{E_e}{E_p} = 10^{-2} \text{ this verifies option (b)}$$

$$\text{Now, } p_e = m_e v_e$$

$$= m_e \times \frac{c}{100} \left(\because v_e = \frac{c}{100} \right)$$

$$\Rightarrow \frac{p_e}{m_e c} = \frac{1}{100}$$

$$= 10^{-2} \text{ this verifies option (c).}$$

12. Photons absorbed in matter are converted to heat. A source emitting n photon sec^{-1} of frequency ν is used to convert 1 kg of ice at 0°C to water at 0°C . Then, the time T taken for the conversion

(a) decreases with increasing n , with ν fixed.

(b) decreases with n fixed, ν increasing

(c) remains constant with n and ν changing such that $n\nu = \text{constant}$.

(d) increases when the product $n\nu$ increases.

Ans: the amount of heat required to convert 1 kg of ice at 0°C to water at 0°C is

$$E = ML$$

$$E' = nh\nu$$

Where, n = no. of photons incident per second.

Let time T is taken by radiation to melt the ice at 0°C . Then,

$$E = nh\nu t$$

$$mL = n h\nu t$$

$$t = \frac{mL}{nh\nu}$$

$$\therefore t \propto \frac{1}{n} \text{ and } T \propto \frac{1}{\nu}$$

$$\text{Or } t \propto \frac{1}{n\nu} [\because m, L \text{ and } h \text{ are constant}]$$

This proves option (a), (b) and (c).

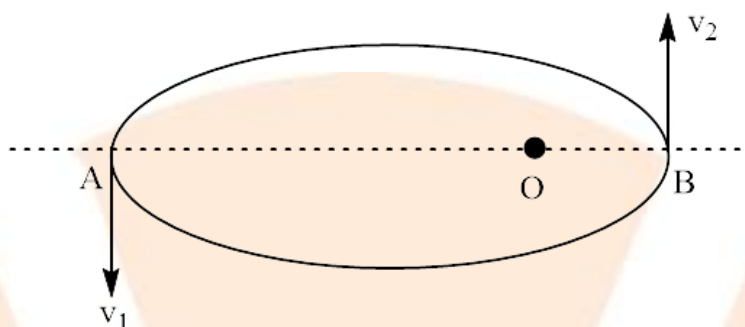
$\therefore$ if $n\nu$ increases, T decreases which does not verify option (d).

13. A particle moves in a closed orbit around the origin, due to a force which is directed towards the origin. The de-Broglie wavelength of the particle varies cyclically between two values λ_1 and λ_2 with $\lambda_1 > \lambda_2$. Which of the following statements are true?

- (a) The particle could be moving in a circular orbit with origin as centre
- (b) The particle could be moving in an elliptic orbit with origin as its focus.
- (c) When the de Broglie wavelength is λ_1 , the particle is nearer the origin than when its value is λ_2
- (d) When the de-Broglie wavelength is λ_2 , the particle is nearer the origin than when its value is λ_1 .

Ans: The de-Broglie wavelength varies regularly between two values λ_1 and λ_2 of the particle. This is possible when the particle is operating in an elliptical

orbit with origin as one of its focuses then. As if λ_1 and λ_2 are equal, then their speed must be equal and particle must move in circular orbit. Hence this verifies option (b).



Let v_1 and v_2 be the speeds of particle at A and B respectively and origin is at O. If particle A and B associated with the de-Broglie wavelengths λ_1 and λ_2 respectively, then

$$\lambda_1 = \frac{h}{mv_1}$$

$$\text{And } \lambda_2 = \frac{h}{mv_2}$$

$$\text{So, } \frac{\lambda_1}{\lambda_2} = \frac{v_2}{v_1} [\because \lambda_1 > \lambda_2]$$

$$\text{Thus, } v_1 < v_2 \dots\dots(i)$$

According to the law of conservation of angular momentum, the speed of the particle will be more when it is closer to focus. This proved the option (d).

Very Short Answer Type Questions

14. A proton and an α -particle are accelerated, using the same potential difference. How are the de-Broglie wavelengths λ_p and λ_α related to each other?

Ans: The both particles are promoted at the same potential difference so their $K.E.s$ will be equal i.e.,

$$K_1 = K_2 = K = qV \left(\because V = \frac{W}{q} \right)$$

$$\text{So, } \lambda = \frac{h}{\sqrt{2mK}}$$

$$\lambda_d = \frac{h}{\sqrt{2mqV}}$$

$$\therefore \frac{\lambda_p}{\lambda_\alpha} = \frac{h}{\sqrt{2m_p q_p V_p}} \times \frac{\sqrt{2m_\alpha q_\alpha V_\alpha}}{h}$$

$$m_\alpha = 4m_p; q_\alpha = 2e; q_p = e; V_p = V_\alpha = V \text{ (P.D. applied)}$$

$$\frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{2 \times 4m_p \cdot 2eV}{2m_p eV}}$$

$$\lambda_p = \sqrt{8}\lambda_\alpha$$

Thus, the de-Broglie wavelength of proton is $\sqrt{8}$ times of alpha (α) particle.

15. (i) In the explanation of photoelectric effect, we assume one photon of frequency ν collides with an electron and transfers its energy. This leads to the equation for the maximum energy $E_{\max}$ of the emitted electron as

$$E_{\max} = h\nu - \phi_0$$

where ϕ_0 is the work function of the metal. If an electron absorbs 2 photons (each of frequency ν) what will be the maximum energy for the emitted electron?

(ii) Why is this fact (two photon absorption) not taken into consideration in our discussion of the stopping potential?

Ans: (i) Here, 2 photons transfer its energy to one electron as

$$E = h\nu$$

$$\therefore E_e = E_p$$

$$h\nu_e = 2h\nu$$

$$\therefore \nu_e = 2\nu$$

Maximum energy of emitted electron is

$$E_{\max} = h\nu_e - \phi_0$$

$$= h(2\nu) - \phi_0$$

$$= 2h\nu - \phi_0$$

(ii) the possibility of absorbing 2 photons by electrons are very low due to their mass difference. So, the probability of such emission of electrons is negligible.

16. There are materials which absorb photons of shorter wavelength and emit photons of longer wavelength. Can there be stable substances which absorb photons of larger wavelength and emit light of shorter wavelength.

Ans: As we know that as the wavelength of a photon increases, its frequency decreases or energy increases.

Case I: Photons which are of shorter wavelength i.e., those which are of larger energy will effuse the photons of smaller energy. Some energy is utilized against work function. So, it is attainable by law of conservation of energy.

Case II: Photons which are of longer wavelength invariably emits photons of shorter wavelength and photons of smaller energy can never effuse photons of larger energy. Some part of energy ($h\nu$) is utilized in work function (ϕ) of metal. Or it will repudiate the law of conservation of energy (universal law). So, it is unendurable in stable materials.

17. Do all the electrons that absorb a photon come out as photo-electrons?

Ans: We can observe that most of the electrons in photoelectric effect pounded by photons are dissipated into the metal by absorbing photons. Thus, some of the energy of photons is absorbed. Generally, one photon cannot emit one electron. Some energy of the photons is consumed against the work function of metal. Thus, some of all electrons which absorb a photon come out from the metal surface.

18. There are two sources of light, each emitting with a power of 100 W . One emits X-rays of wavelength 1 nm and the other visible light at 500 nm . Find the ratio of the number of photons of X-rays to the photons of visible light of the given wavelength?

Ans: Let the energies E_x and E_v given by one photon in X-rays and visible rays then,

$$E_x = h\nu_x$$

And $E_v = h\nu_v$

Let the number of photons n_x and n_v X-rays and visible light to give equal energies as both sources (X-rays and visible) emitting same power 100 W each so

$$n_x h\nu_x = n_v h\nu_v$$

$$\frac{n_x}{n_v} = \frac{\nu_v}{\nu_x}$$

$$\Rightarrow \frac{\frac{1}{\lambda_v}}{\frac{1}{\lambda_x}} = \frac{\lambda_x}{\lambda_v} \left(\because \nu = \frac{c}{\lambda} \text{ or } \nu = \frac{1}{\lambda} \right)$$

$$\frac{n_x}{n_v} = \frac{1 \text{ nm}}{500 \text{ nm}} \quad (\because \lambda_x = 1 \text{ nm and } \lambda_v = 500 \text{ nm})$$

$$n_x : n_v = 1 : 500$$

Short Answer Type Questions

19. Consider Fig.11.1 for photo-emission. How would you reconcile with momentum conservation? Note light (photons) have momentum in a different direction than the emitted electrons.

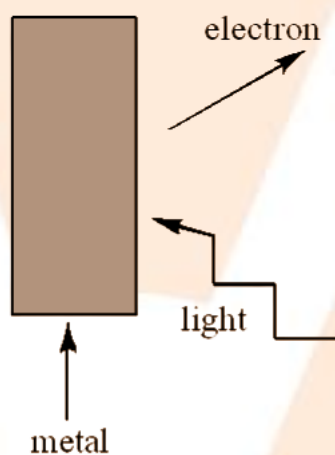


Fig.11.1

Ans: The transfer of momentum of photons to atoms of metal when photons pound to the metal surface by reducing its own speed up to zero. Photons convey this momentum to the nucleus and electrons of the metal. The direction of emission of excited electrons is roughly contrary to the direction of photons. The total momentum transferred by the photons will be equivalent to the momentum of all electrons and nucleus.

20. Consider a metal exposed to light of wavelength 600 nm . The maximum energy of the electron doubles when light of wavelength 400 nm is used. Find the work function in eV .

Ans: let K_1 and K_2 are the maximum energies of emitted electrons when 600 nm and 400 nm visible light is used according to the question

$$K_2 = 2K_1$$

$$K_{\max} = h\nu - \phi$$

$$= \frac{hc}{\lambda} - \phi$$

$$k_1 = \frac{hc}{\lambda_1} - \phi$$

$$k_2 = \frac{hc}{\lambda_2} - \phi$$

$$= 2K_1$$

$$\frac{hc}{\lambda_2} - \phi = 2 \left[\frac{hc}{\lambda_1} - \phi \right]$$

$$= \frac{2hc}{\lambda_1} - 2\phi$$

$$\phi = hc \left[\frac{2}{\lambda_1} - \frac{1}{\lambda_2} \right] (\because hc = 1240\text{ eV nm})$$

$$\therefore \phi = 1240 \left[\frac{2}{600} - \frac{1}{400} \right] \text{ eV}$$

$$= \frac{1240}{200} \left[\frac{2}{3} - \frac{1}{2} \right]$$

$$= 6.2 \frac{(4-3)}{6}$$

The work function $\phi = \frac{6.2}{6}$

$$= 1.03\text{ eV}$$

21. Assuming an electron is confined to a 1nm wide region, find the uncertainty in momentum using Heisenberg Uncertainty principle ($\Delta x \times \Delta p \cong h$). You can assume the uncertainty in position Δx as 1nm. Assuming $p \approx \Delta p$, find the energy of the electron in electron volts.

Ans: As we know that electron revolves in circular path

Thus, $\Delta r = 1 \text{ nm}$

$$= 10^{-9} \text{ m}$$

$$\Delta x \times \Delta p \cong h \text{ [Given]}$$

$$\Delta p = \frac{h}{\Delta x}$$

$$= \frac{6.62 \times 10^{-34}}{2\pi \Delta r} \text{ JS}$$

$$\therefore \Delta p = \frac{6.62 \times 10^{-34}}{2 \times 3.14 \times 10^{-9}} \text{ kg m s}^{-1}$$

$$\Rightarrow \Delta p = \frac{331}{314} \times 10^{-25}$$

$$\therefore E = \frac{1}{2} mv^2$$

$$= \frac{m^2 v^2}{2m}$$

$$= \frac{\Delta p^2}{2m}$$

$$= \frac{331 \times 331 \times 10^{-50+31}}{314 \times 314 \times 18.2} \text{ J}$$

$$= \frac{331 \times 331 \times 10^{-19} \times 1.6 \times 10^{-19}}{314 \times 314 \times 18.2} e$$

$$= \frac{331 \times 331 \times 16}{314 \times 314 \times 182}$$

$$= 3.8 \times 10^{-2} \text{ eV}$$

22. Two monochromatic beams A and B of equal intensity I, hit a screen. The number of photons hitting the screen by beam A is twice that by beam B. Then what inference can you make about their frequencies?

Ans: Given,

$$I_A = I_B$$

$$n_A h\nu_A = n_B h\nu_B$$

$$\therefore n_A = 2n_B$$

$$\therefore 2n_B \nu_A = n_B \nu_B$$

$$\Rightarrow 2\nu_A = \nu_B$$

Thus, the frequency of the source B is twice of the frequency of source A.

23. Two particles A and B of de Broglie wavelengths λ_1 and λ_2 combine to form a particle C. The process conserves momentum. Find the de-Broglie wavelength of the particle C. (The motion is one dimensional).

Ans: According to the de-Broglie wavelengths

$$\lambda = \frac{h}{p}$$

$$\text{Or } p = \frac{h}{\lambda}$$

$$\Rightarrow p_1 = \frac{h}{\lambda_1}, p_2 = \frac{h}{\lambda_2} \text{ and } p_3 = \frac{h}{\lambda_3}$$

$$p_1 + p_2 = p_3$$

$$\frac{h}{\lambda_1} + \frac{h}{\lambda_2} = \frac{h}{\lambda_3} \quad (\lambda_3 = \text{the wavelength of particle C})$$

$$\frac{1}{\lambda_1} + \frac{1}{\lambda_2} = \frac{1}{\lambda_3}$$

$$\frac{\lambda_2 + \lambda_1}{\lambda_1 \lambda_2} = \frac{1}{\lambda_3}$$

Case I: When p_1 and p_2 are the positive then

$$\lambda_3 = \frac{\lambda_2 \lambda_1}{\lambda_1 + \lambda_2}$$

Case II: When both are negative then

$$\lambda_3 = \left| \frac{-\lambda_2 \lambda_1}{\lambda_1 + \lambda_2} \right| = \frac{\lambda_2 \lambda_1}{\lambda_1 + \lambda_2}$$

Case III: $p_A > 0$, $p_B > 0$

$$\frac{h}{\lambda_3} = \frac{h}{\lambda_1} - \frac{h}{\lambda_2}$$

$$\text{Or } \frac{1}{\lambda_3} = \frac{\lambda_2 - \lambda_1}{\lambda_1 \lambda_2}$$

$$\lambda_3 = \frac{\lambda_1 \lambda_2}{\lambda_2 - \lambda_1}$$

Case IV: $p_A < 0$, $p_B > 0$

$$\frac{h}{\lambda_3} = \frac{h}{\lambda_2} - \frac{h}{\lambda_1}$$

$$\Rightarrow \frac{h}{\lambda_3} = \frac{(\lambda_1 - \lambda_2)h}{\lambda_1 \lambda_2}$$

$$\therefore \lambda_3 = \frac{\lambda_1 \lambda_2}{\lambda_1 - \lambda_2}$$

24. A neutron beam of energy E scatters from atoms on a surface with a spacing $d = 0.1 \text{ nm}$. The first maximum of intensity in the reflected beam occurs at $\theta = 30^\circ$. What is the kinetic energy E of the beam in eV ?

Ans: According to the Bragg's law of diffraction, condition for n th maxima is

$$2d \sin \theta = n\lambda$$

Where, $n = 1$,

$$\text{So, } \lambda = 2d \sin \theta$$

Where $\theta = 30^\circ$ (Given)

$$= 2 \times 0.1 \times 10^{-9} \sin 30^\circ (\because d = 0.1 \text{ nm})$$

$$p = \frac{h}{\lambda}$$

$$= \frac{6.6 \times 10^{-34}}{10^{-10}}$$

$$= 6.6 \times 10^{-24} \text{ kg m s}^{-1}$$

$$E = \frac{1}{2} mv^2$$

$$= \frac{1}{2} \frac{m^2 v^2}{m}$$

$$= \frac{p^2}{2m}$$

$$E = \frac{6.6 \times 6.6 \times 10^{-48}}{2 \times 1.6 \times 10^{-27}} \text{ J}$$

$$= \frac{66 \times 66 \times 10^{-48+46}}{2 \times 16 \times 16}$$

$$E = \frac{33 \times 33}{126} \times 10^{-2}$$

$$= 8.5 \times 10^{-2}$$

$$= 0.085 \text{ eV}$$

Long Answer Type Questions

25. Consider a thin target (10^{-2} m square, 10^{-3} m thickness) of sodium, which produces a photo-current of $100 \mu\text{A}$ when a light of intensity 100 W m^{-2} ($\lambda = 660 \text{ nm}$) falls on it. Find the probability that a photo-electron is produced when a photon strikes a sodium atom.

[Take density of Na = 0.97 kgm^{-3}].

Ans: Area of the square sheet (A) = $10^{-2} \times 10^{-2}$

$$= 10^{-4} \text{ m}^2$$

Thickness (d) = 10^{-3} m

Current (i) = $100 \mu\text{A} = 10^{-4} \text{ A}$

Intensity (I) = 100 W m^{-2}

Mass of target (m) = $\text{Volume} \times \text{density}$

$$m = \text{area of sheet} \times \text{thickness} \times \text{density}$$

$$\Rightarrow m = (10^{-4} \times 10^{-3}) \times 0.97 \text{ kg}$$

$$\Rightarrow m = 0.97 \times 10^{-7} \text{ kg}$$

$$\Rightarrow m = 0.97 \times 10^{-4} \text{ gm}$$

$$\therefore \text{number of Na atoms in target} = \frac{6.023 \times 10^{23}}{23} \times 0.97 \times 10^{-4}$$

$$= 0.254 \times 10^{19}$$

$$\text{Number of Na atoms in Na target} = 2.54 \times 10^{18} \text{ atoms}$$

$$\text{Total energy falling per second on target} = nh\nu$$

$$\text{Intensity} \times \text{Area} = n \times h \times \frac{c}{\lambda}$$

$$I \times A = \frac{nhc}{\lambda}$$

$$n = \frac{IA\lambda}{hc}$$

$$= \frac{100 \times 10^{-4} \times 660 \times 10^{-9}}{6.6 \times 10^{-34} \times 3 \times 10^8}$$

$$= \frac{1000 \times 660}{66 \times 3} \times 10^{-13-8+34}$$

$$= \frac{10000}{3} \times 10^{-24+34}$$

$$= \frac{10}{3} \times 10^3 \times 10^{13}$$

$$\text{Number of photons (n) incident per second on Na metal} = 3.3 \times 10^{16}$$

Assuming P is the possibility of emission of photo-electrons per atom per photon.

Number of photo-electrons emitted per second (N) = $P \cdot n$ (here n is no. of sodium atoms)

$$N = P \times 3.3 \times 10^{16} \times 2.54 \times 10^{18}$$

$$i = 100 \mu A = 10^{-4} A = Ne$$

$$\Rightarrow N = \frac{i}{e}$$

$$P = \frac{i}{e \times 3.3 \times 10^{16} \times 2.54 \times 10^{18}}$$

$$P = \frac{10^{-4}}{1.6 \times 10^{-19} \times 3.3 \times 10^{16} \times 2.54 \times 10^{18}}$$

$$= \frac{10^{-4-34+19}}{13.4}$$

$$0.075 \times 10^{-19}$$

$$P = 7.5 \times 10^{21}$$

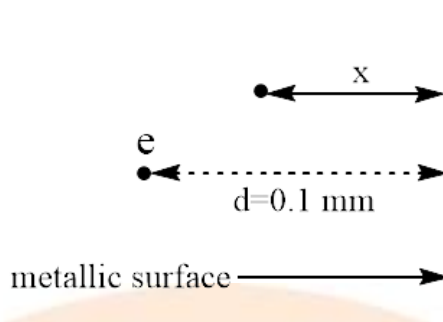
26. Consider an electron in front of metallic surface at a distance d (treated as an infinite plane surface). Assume the force of attraction by the plate is given as $\frac{1}{4} \cdot \frac{q^2}{4\pi\epsilon_0 d^2}$. Calculate work in taking the charge to an infinite distance from the plate. Taking $d = 0.1\text{nm}$, find the work done in electron volts. [Such a force law is not valid for $d < 0.1\text{nm}$].

Ans: According to the question

$$F = \frac{1}{4} \cdot \frac{q^2}{4\pi\epsilon_0 d^2}$$

Assuming that at any instant electron is at distance x from the metal surface. Attractional force between metal surface and electron is F

$$F = \frac{1}{4} \cdot \frac{q^2}{4\pi\epsilon_0 x^2}$$



Work done by external in taking the electron from distance d to ∞ is

$$W.D. = \int_d^{\infty} F \cdot dx$$

$$\Rightarrow W.D. = \int_d^{\infty} \frac{q^2 dx}{4 \times 4\pi\epsilon_0 x^2}$$

$$= \frac{q^2}{4 \times 4\pi\epsilon_0} \int_d^{\infty} x^{-2} dx$$

$$= \frac{q^2}{4 \times 4\pi\epsilon_0} \left[\frac{x^{-1}}{-1} \right]_d^{\infty}$$

$$W.D. = -\frac{q^2}{4.4\pi g e_0} \left[\frac{1}{x} \right]_d^{\infty}$$

$$= \frac{-q^2 k}{4} \left[\frac{1}{\infty} - \frac{1}{d} \right]$$

$$W.D. = \frac{+kq^2}{4d} \left[d = 0.1 \text{ nm} = 10^{-10} \text{ m} \right]$$

Work done here is positive

$$\text{So, } W.D. = \frac{(1.6 \times 10^{-19})^2 \times 9 \times 10^9}{4 \times 10^{-10}} \text{ J}$$

$$= \frac{1.6 \times 9 \times 1.6 \times 10^{-38+9+10}}{4 \times 1.6 \times 10^{-19}} \text{ eV}$$

$$= \frac{9 \times 1.6 \times 10^{-19+19}}{4}$$

$$= 3.6 \text{ eV}$$

27. A student performs an experiment on photoelectric effect, using two materials A and B. A plot of V_{stop} vs ν is given in Fig. 11.2.

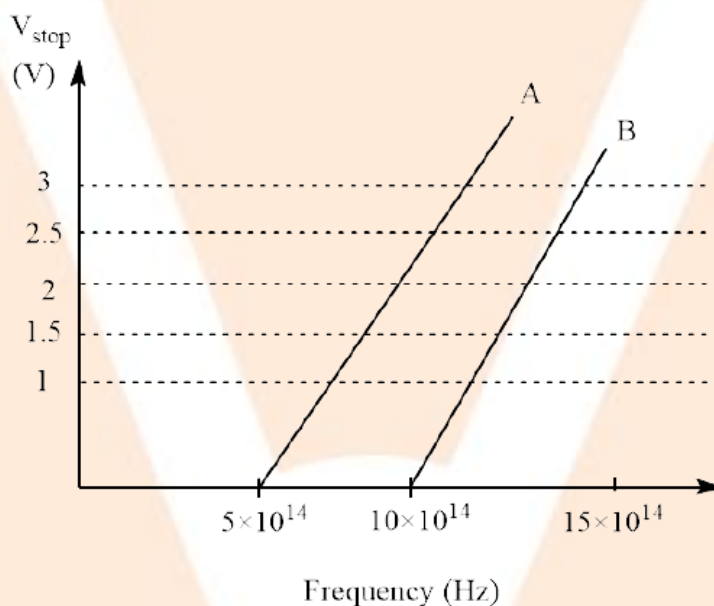


Fig.11.2

(i) Which material A or B has a higher work function?

(ii) Given the electric charge of an electron = $1.6 \times 10^{-19} \text{ C}$, find the value of h obtained from the experiment for both A and B. Comment on whether it is consistent with Einstein's theory:

Ans: (i) As we know

$$\phi_0 = h\nu_0$$

Where ν_0 = Threshold frequency

h = Planck's constant

For metal A $\nu_A = 5 \times 10^4 \text{ Hz}$

For metal B $\nu_B = 10 \times 10^4 \text{ Hz}$

$$\frac{\phi_A}{\phi_B} = \frac{h\nu_A}{h\nu_B}$$

$$= \frac{5 \times 10^{14}}{10 \times 10^{14}}$$

$$\therefore \phi_B = 2\phi_A$$

Thus, work function of material B is twice of material A.

(ii) By the differentiation of potential

$$V = \frac{W}{Q}$$

If $W = E$ and $Q = e$ (charge on an electron)

Then,

$$V = \frac{E}{e}$$

$$\text{Or } E = eV$$

$$h\nu = eV \quad (\because E = h\nu)$$

Differentiating both sides, we get

$$h.d\nu = e.dV$$

$$h = e \cdot \frac{dV}{d\nu}$$

For metal A,

$$h = \frac{1.6 \times 10^{-19} [2 - 0]}{(10 - 5) \times 10^{14}}$$

$$= \frac{3.2}{5} \times 10^{-19+14}$$

$$= 0.64 \times 10^{-33} \text{ JS}$$

$$= 6.4 \times 10^{-33} \text{ JS}$$

$$h = 6 \times 10^{-34} \text{ JS} \dots\dots\dots(I)$$

For metal B,

$$h = \frac{e \times (2.5 - 0)}{(10 - 5) \times 10^{14}}$$

$$= \frac{1.6 \times 10^{-19} \times 2.5 \times 10^{-14}}{5}$$

$$= \frac{4.00}{5} \times 10^{-33}$$

$$= 0.8 \times 10^{-33}$$

$$h = 8 \times 10^{-34} \text{ JS} \dots\dots(II)$$

The Planck's constant (h)'s value for both experimental graphs are not equal, so the experiment is not consistent with Einstein's theory.

But because of experimental limitation values are very near to $6.6 \times 10^{-34} \text{ JS}$, so can be considered consistent with Einstein theory.

28. A particle A with a mass m_A is moving with a velocity v and hits a particle B (mass m_B) at rest (one dimensional motion). Find the change in the de-Broglie wavelength of the particle A. Treat the collision as elastic.

Ans: Because the collision is elastic, the law of conservation of momentum and kinetic energy are followed.

$$m_A v + m_B (0) = m_A v_1 + m_B v_2$$

$$m_A (v - v_1) = m_B v_2 \dots (I)$$

$$\text{And } \frac{1}{2} m_A v^2 + \frac{1}{2} m_B (0)^2 = \frac{1}{2} m_A v_1^2 + \frac{1}{2} m_B v_2^2$$

$$m_A (v^2 - v_1^2) = m_B v_2^2$$

$$m_A (v - v_1)(v + v_1) = m_B v_2^2 \dots (II)$$

On dividing (II) by (I) we get,

$$v + v_1 = v_2 \dots (III)$$

$$v = v_2 - v_1$$

Putting (III) in (I)

$$m_A v - m_A v_1 = m_B (v + v_1)$$

$$m_A v - m_A v_1 = m_B v + m_B v_1$$

$$m_A v - m_B v = m_A v_1 + m_B v_1$$

$$(m_A - m_B) v = (m_A + m_B) v_1$$

$$v_1 = \frac{(m_A - m_B)}{(m_A + m_B)} v$$

From (II),

$$v_2 = v + \frac{(m_A - m_B)}{(m_A + m_B)} v$$

$$= v \left[1 + \frac{(m_A - m_B)}{(m_A + m_B)} \right]$$

$$= v \left[\frac{(m_A + m_B + m_A - m_B)}{(m_A + m_B)} \right]$$

$$= \frac{2m_A v}{(m_A + m_B)}$$

$$\lambda_{A \text{ initial}} = \frac{h}{m_A v}$$

$$\lambda_{A \text{ final}} = \frac{h}{m_A v}$$

$$\lambda_{A \text{ final}} = \frac{h(m_A + m_B)}{m_A(m_A - m_B)v}$$

$$\Delta\lambda = \lambda_{A \text{ final}} - \lambda_{A \text{ initial}}$$

$$= \frac{h}{m_A v} \left[\frac{(m_A + m_B)}{(m_A - m_B)} - 1 \right]$$

$$= \frac{h}{m_A v} \left[\frac{(m_A + m_B) - (m_A - m_B)}{(m_A - m_B)} \right]$$

Thus, the change in the de-Broglie wavelength $\Delta\lambda = \frac{2m_B h}{m_A(m_A - m_B)v}$

29. Consider a 20 W bulb emitting light of wavelength $5000 \text{ \AA}$ and shining on a metal surface kept at a distance 2m . Assume that the metal surface has work function of 2 eV and that each atom on the metal surface can be treated as a circular disk of radius $1.5 \text{ \AA}$.

(i) Estimate no. of photons emitted by the bulb per second. [Assume no other losses]

(ii) Will there be photoelectric emission?

(iii) How much time would be required by the atomic disk to receive energy equal to work function (2 eV)?

(iv) How many photons would an atomic disk receive within the time duration calculated in (iii) above?

(v) Can you explain how the photoelectric effect was observed instantaneously?

Ans: Given, $P = 20 \text{ W}$, $\lambda = 5000 \text{ \AA} = 5000 \times 10^{-10} \text{ m}$, $r = 1.5 \text{ \AA} = 1.5 \times 10^{-10} \text{ m}$ (atomic radius) and $\phi = 2 \text{ eV}$

Assuming the number of photons emitted by bulb per second is n_1 then Power P is

$$P = n_1 h \nu$$

$$\text{Or } P = n_1 \frac{c}{\lambda}$$

$$\therefore n_1 = \frac{P \lambda}{hc}$$

$$= \frac{20 \times 5000 \times 10^{-10}}{6.62 \times 10^{-34} \times 3 \times 10^8}$$

$$n_1 = \frac{100000 \times 10^{-10-8+34}}{19.86} \approx \frac{100000}{20} \times 10^{-18+34}$$

$$n_1 = 5 \times 10^{16+3}$$

$$= 5 \times 10^{19} \text{ (no. of photons per second)}$$

Number of photons effused by bulb per second

$$n_1 = 5 \times 10^{19}$$

(ii) incident photon's energy

$$E = h \nu$$

$$= \frac{hc}{\lambda}$$

$$= \frac{6.62 \times 10^{-34} \times 3 \times 10^8}{5000 \times 10^{-10}} J$$

$$= \frac{19.86 \times 10^{-34+10+8}}{5000 \times 1.6 \times 10^{-19}} eV$$

$$= 2.48 eV$$

Energy of photon = $h\nu$

$$= \frac{hc}{\lambda}$$

$$= \frac{(6.62 \times 10^{-34})(3 \times 10^8)}{5000 \times 10^{-10}} J$$

$$\Rightarrow \frac{6.62 \times 3 \times 10^{-24+8}}{5000 \times 1.6 \times 10^{-19}} eV = \frac{19.86 \times 10^{-16+19}}{8000}$$

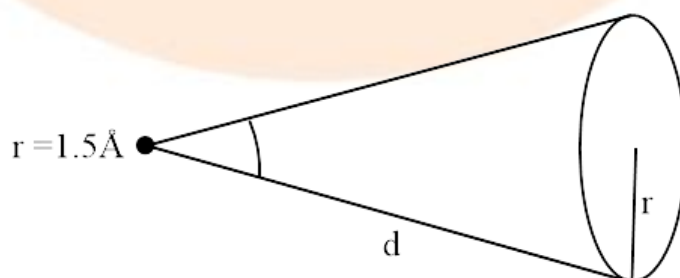
$$\text{Energy of photon} = \frac{20 \times 10^3}{8 \times 10^3}$$

$$= \frac{5}{2} eV$$

$$= 2.5 eV$$

The photoelectric emission happened because the energy of an incident photon is comparably more than $2 eV$ i.e., metal surface's work function.

(iii) Assuming ΔT time depleted in getting the energy ϕ (work function of metal) energy sustained by atomic disk in



At time Δt ,

$$E = P \times A \cdot \Delta t$$

Energy transmitted by bulb in full solid angle $4\pi d^2$ to atoms $= 4\pi d^2 \phi$

$$\therefore p \times \pi r^2 \Delta t = 4\pi d^2 \phi$$

$$\Delta t = \frac{4d^2 \phi}{P r^2} = \frac{4 \times 2 \times 2 \times 2 \times 1.6 \times 10^{-19}}{20 \times 1.5 \times 1.5 \times 10^{-10} \times 10^{-10}} \text{ sec}$$

$$\Delta t = \frac{12.8 \times 10^{-19+20}}{5 \times 2.25}$$

$$= \frac{128}{12.25}$$

$$= 11.4 \text{ sec}$$

(iv) Number of photons acquired by one atomic disk in time Δt

$$N = \frac{n_1 r^2 \Delta t}{4\pi d^2}$$

$$= \frac{n_1 r^2 \Delta t}{4d^2}$$

$$= \frac{5 \times 10^{19} \times 1.5 \times 1.5 \times 10^{-20} \times 11.4}{4 \times 2 \times 2} [n_1 \text{ from part (i) from part (iii)}]$$

$$N = \frac{12.25 \times 11.4 \times 10^{-1}}{8 \times 2 \times 2} \approx 0.80 \cong 1 \text{ photon per atom}$$

$$N = 1 \text{ photon per atom .}$$

(v) 11.4 sec is the time of emission of electrons. Thus, the photoelectric emission is not immediate in the problem. Thus, it takes about 11.4 sec. There is a collision among photon and free electron and nucleus in photoelectric which lasts for very short time interval (10^{-9} sec) therefore we say photoelectric emission is expeditious.